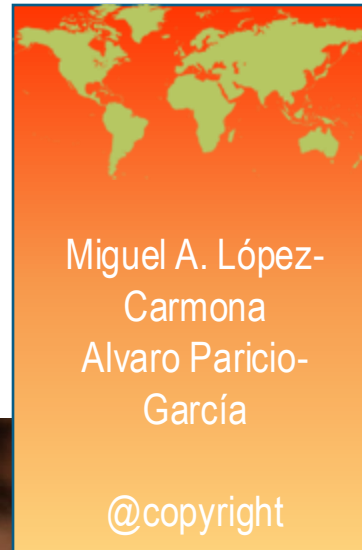




SCM key concepts



Co-funded by the
European Union

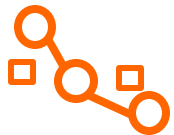


Universidad
de Alcalá

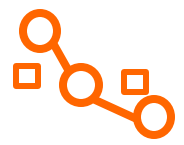


Four questions

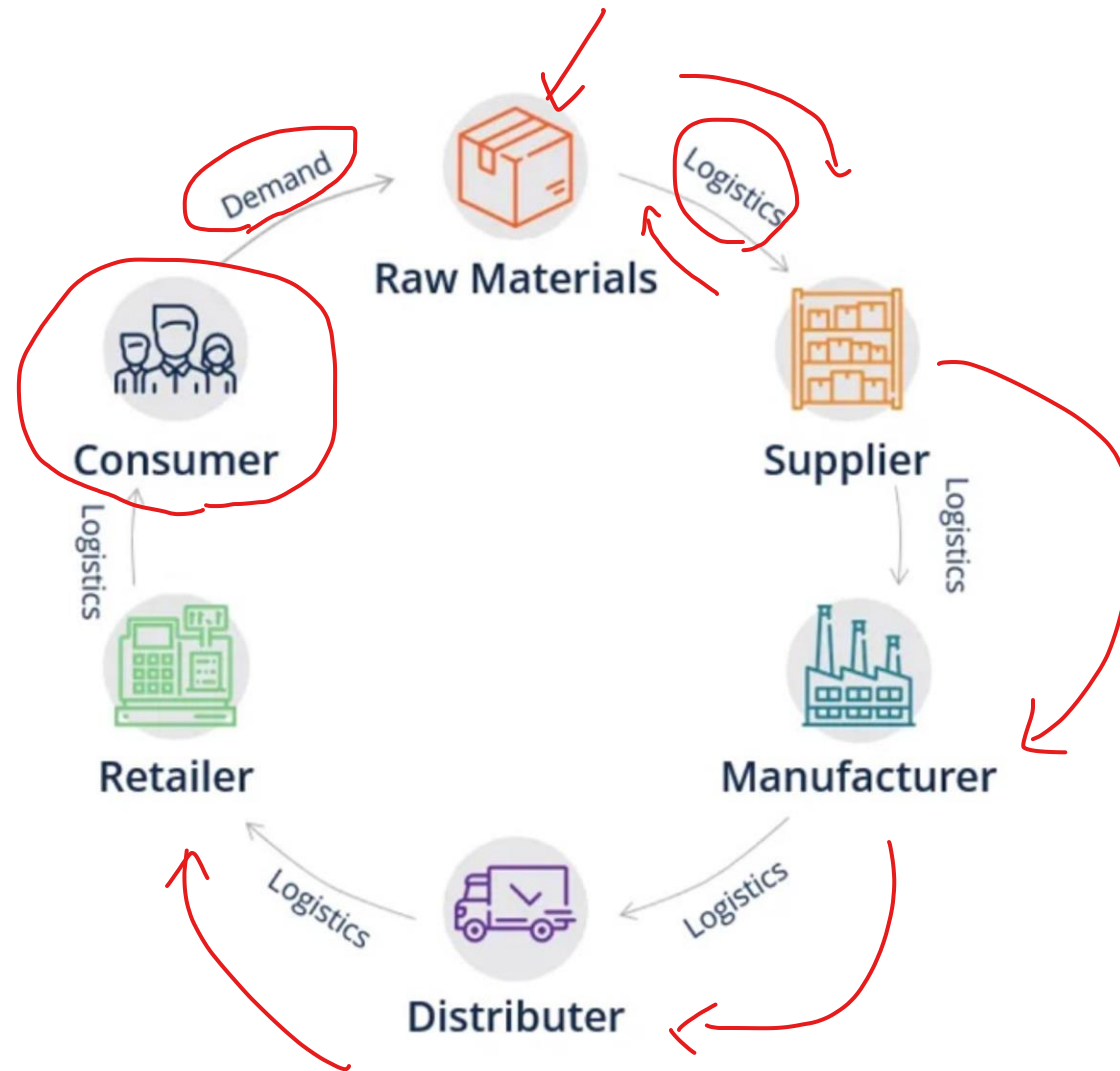
1. What is a supply chain (SC)?
2. What is logistics and supply chain management?
3. What should the SC look like?
4. What are the challenges?

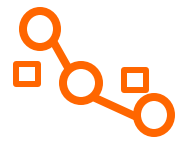


What is a Supply Chain?

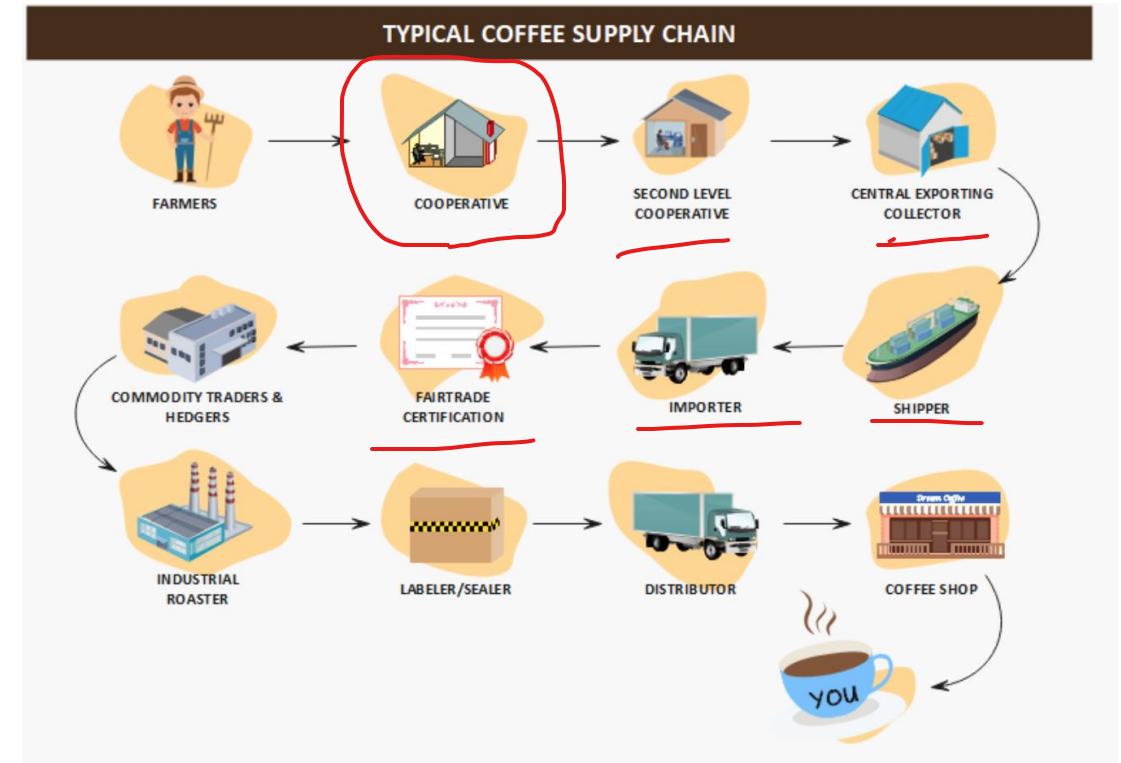
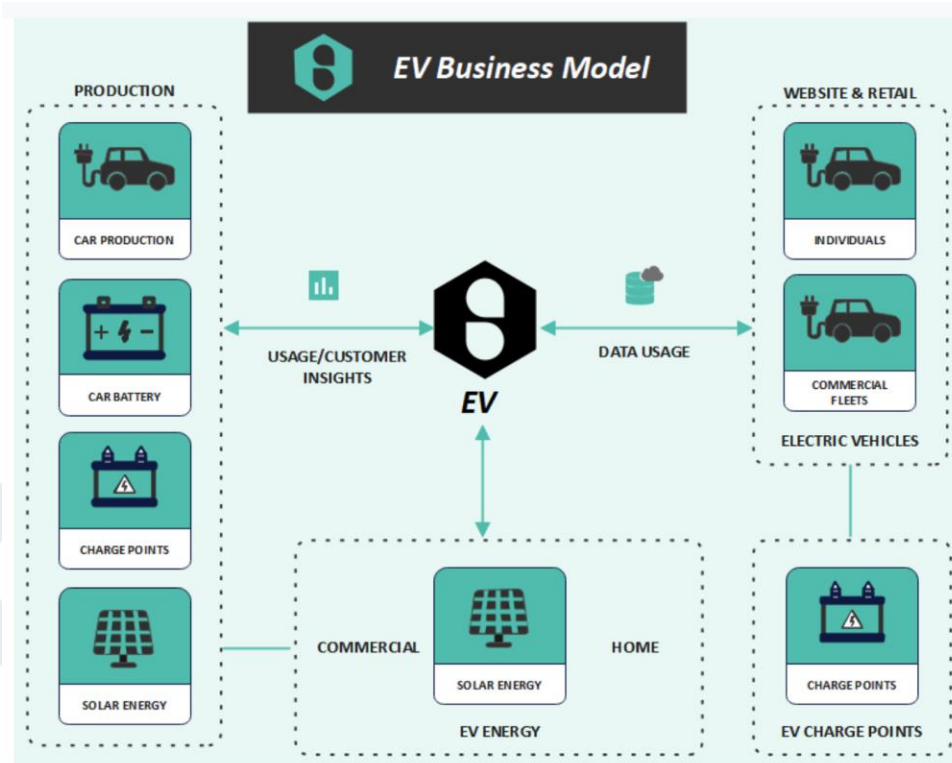


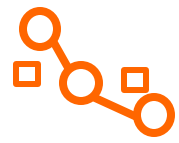
What is a supply chain?





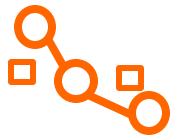
What is a supply chain?



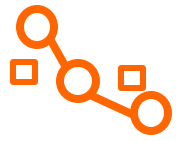


What is a supply chain?

- The main objective of a SC is to satisfy the customer's needs.
 - Only one source of revenue: the customer.
 - Payments between parties are only exchanges of funds.
 - The division of payments within the SC is a function of power, market conditions, etc.
- Supply chains try to maximize the total value generated
 - = [What the customer pays] - [Total effort invested]



Logistics vs. Supply Chain Management



Logistics versus supply chain management

Logistics involves . . . "managing the flow of goods, information, cash
and ideas through the coordination of supply chain processes and through the
strategic addition of place, period and pattern values".

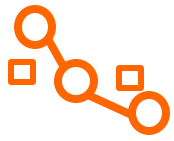
MIT Center for Transportation and Logistics

"Supply Chain Management is concerned with the management of materials,
information and financial flows in a network consisting of suppliers,
manufacturers, distributors and customers."

Stanford Supply Chain Forum

"Call it distribution, logistics or supply chain management. By whatever name, it's the winding,
rough and cumbersome process by which companies move materials, parts and products to
customers."

Fortune (1994)

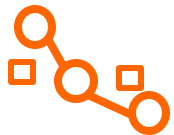


Logistics versus supply chain management

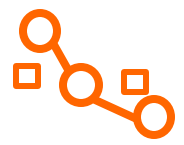
According to the Council of Supply Chain Management Professionals. . .

Logistics management is the part of supply chain management that plans, implements and controls the efficient and effective forward and reverse flow and storage of goods, services and related information between the point of origin and the point of consumption to meet customer requirements.

Supply chain management encompasses the planning and management of all activities involved in sourcing and procurement, conversion and all logistics management activities. Importantly, it also includes coordination and collaboration with channel partners, which can be suppliers, intermediaries, external service providers and customers. In essence, supply chain management integrates supply and demand management within and between companies.



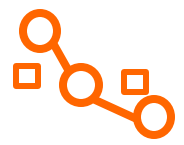
Supply Chain Management Perspectives



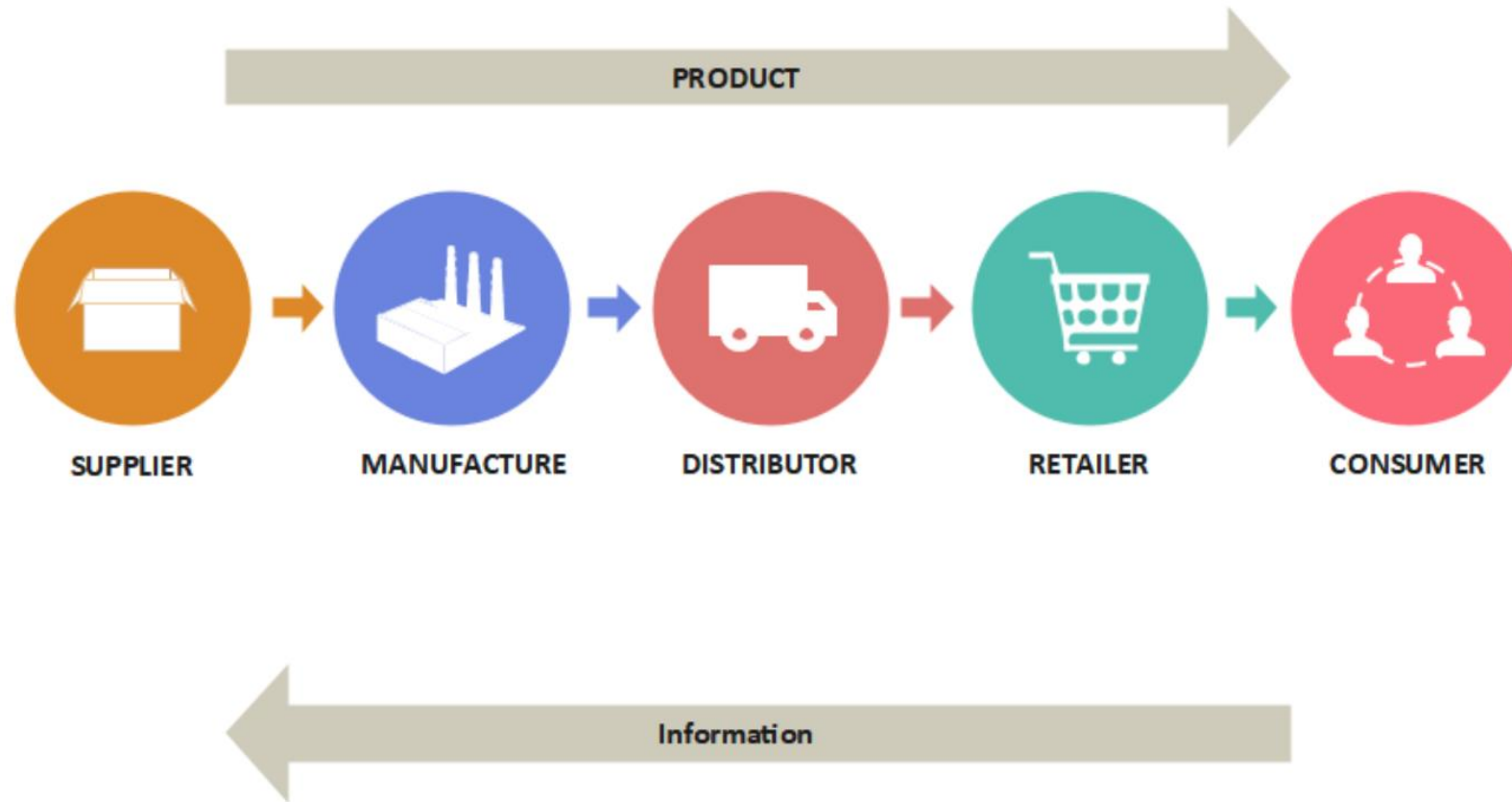
Supply Chain Perspectives

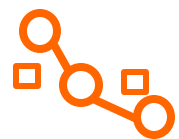


Image source: Arntzen, B. MIT Center for Transportation and Logistics, Hi-Viz Research Project (2013).

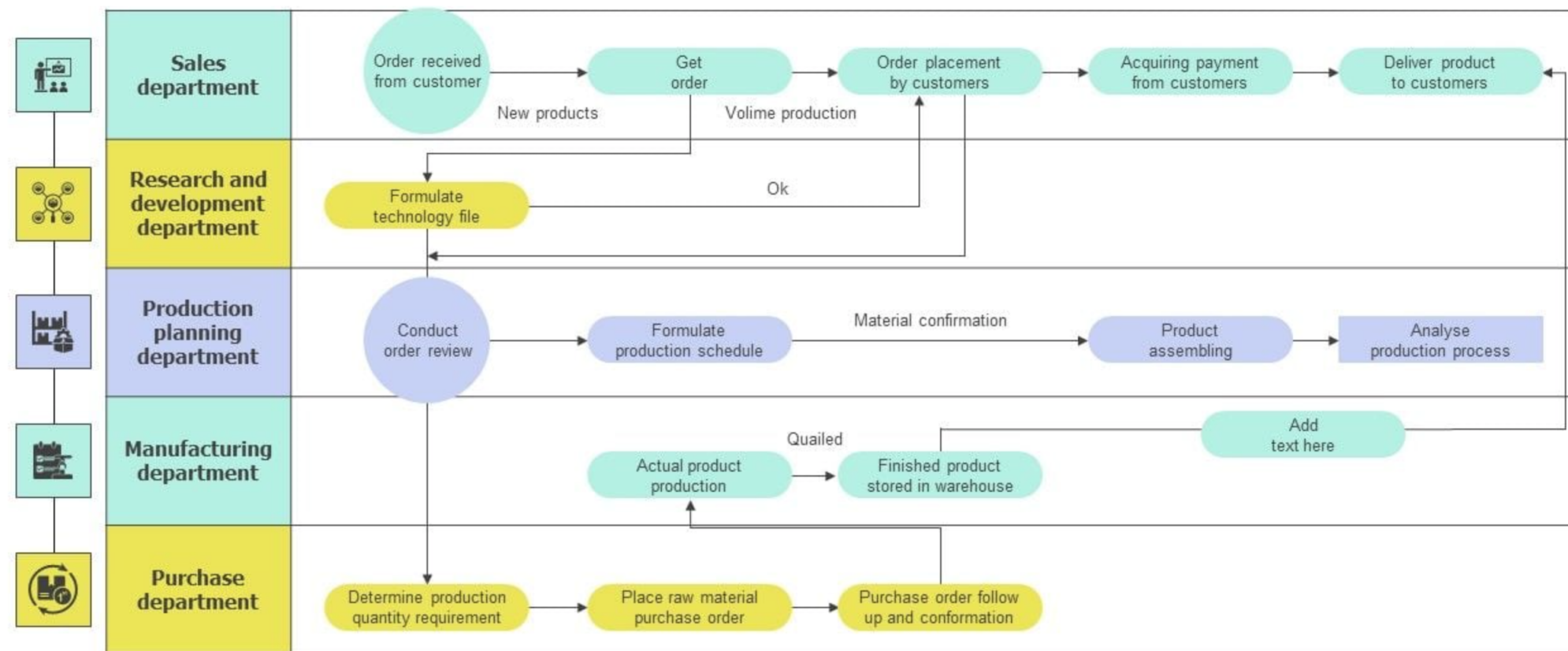


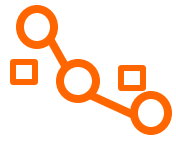
Basic SC perspective



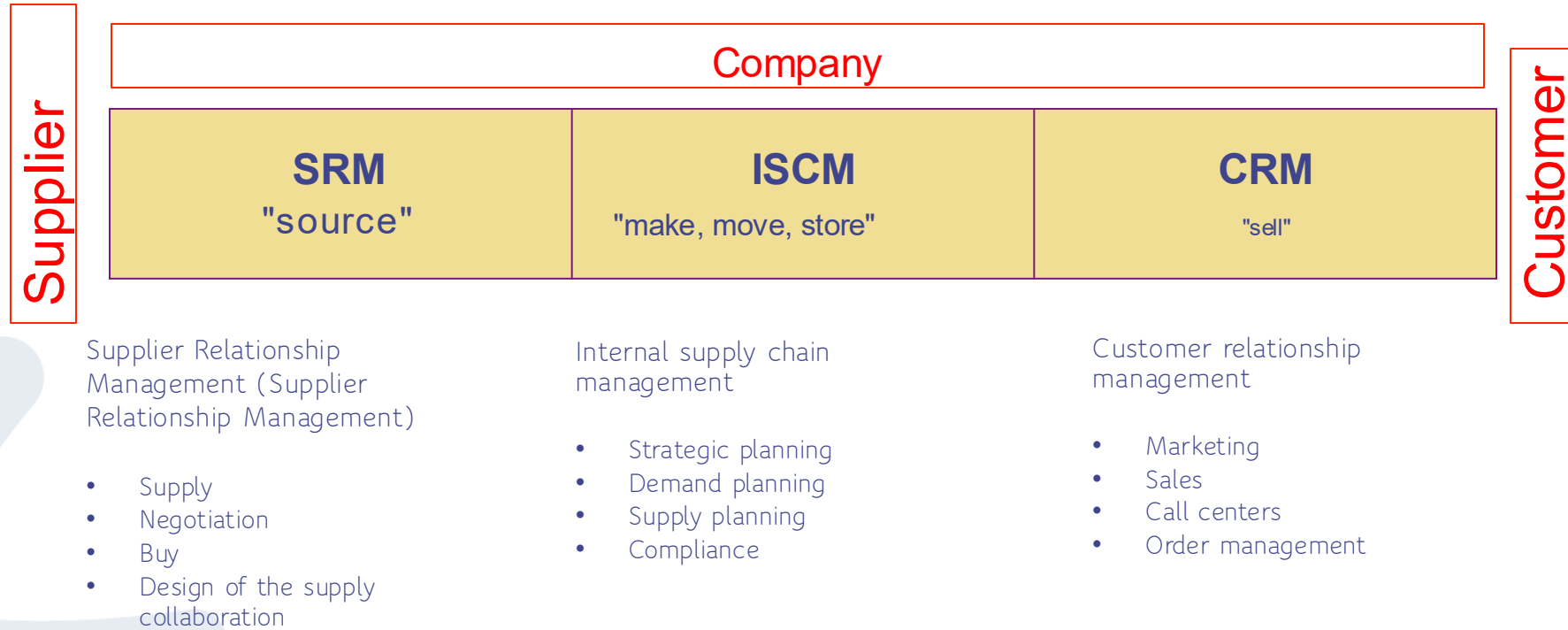


SC process view





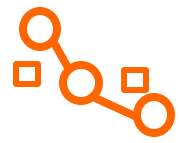
Software Perspective



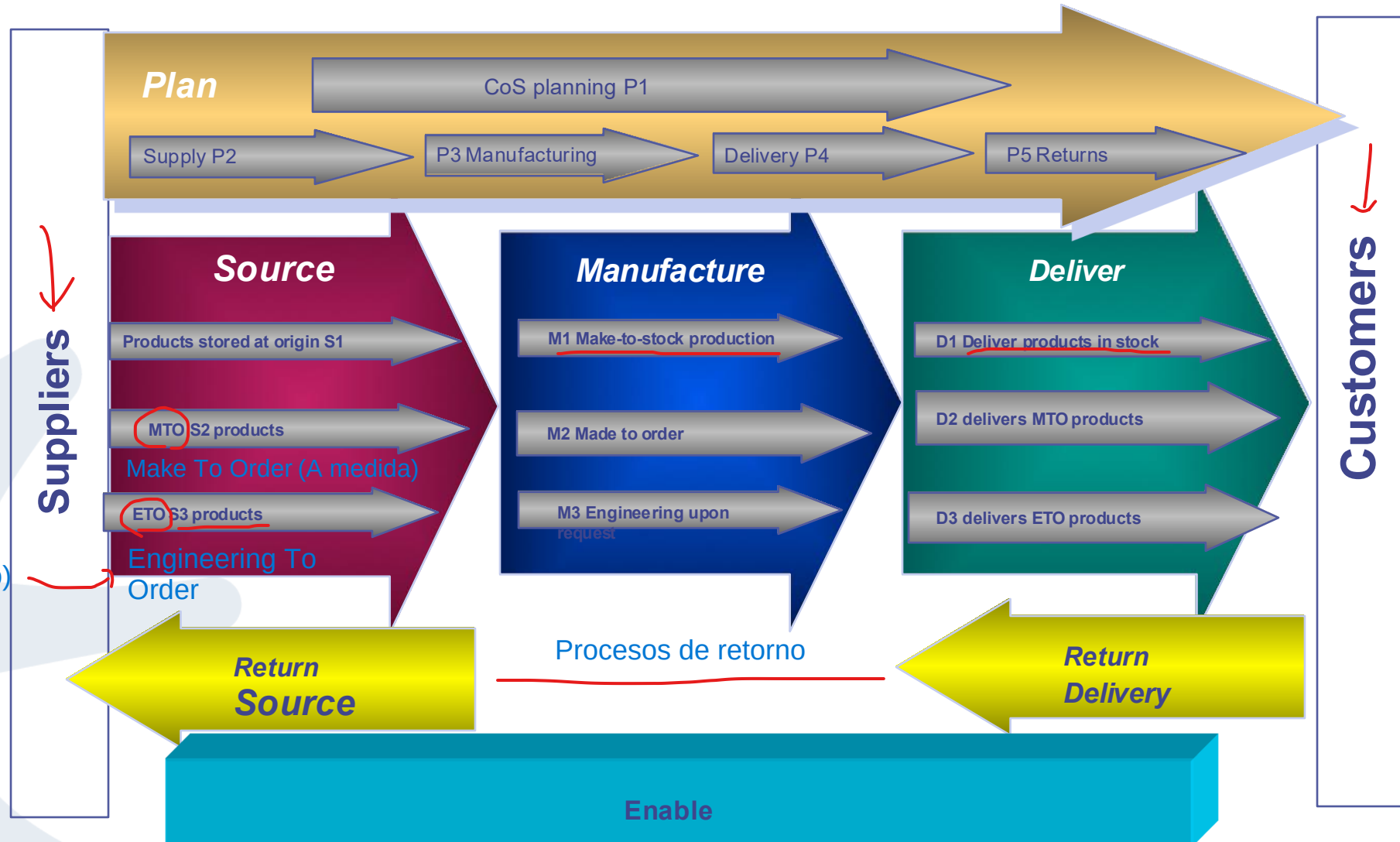
Different macro-processes must be aligned

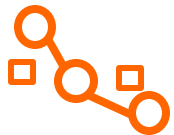
They belong to different groups (Marketing-CRM, Manufacturing-ISCM, Purchasing-SRM).

Forecasts, schedules, design, etc. must be coordinated.

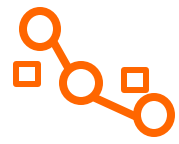


Supply Chain Operations Reference Model (SCOR)



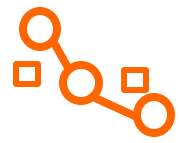


What are the main challenges?



Why is it so difficult?

- Metrics: How is a system measured?
- Politics and player power: who wins?
 - Family store vs. megastores
 - Mega-retailers versus mega-manufacturers
- Visibility: who can see what and how quickly?
 - The data is stored separately.
 - Not all parties have equal access to data
 - Massive data $\neq$ Shared and accessible information.



Why is it so difficult?

- Uncertainty - Who knows what will happen?
 - Variable product demand (shorter life cycles)
 - Variable manufacturing performance
 - Unreliable supply of raw materials
 - Inconsistent transit delivery times
- Increased complexity: why is it becoming more difficult?
 - Explosive number of stock keeping units (SKU)
 - Higher and diverging customer demands
 - New and merged channels (Omni-Channel)
- Global operations: why do we never close?
 - Most of the companies supply and sell worldwide.
 - Multiple regions, time zones, languages and cultures